Xian **WANG**

PhD Candidate in Human-Computer Interaction · XR Systems, Human-Robot Interaction, and Human-Centered AI

Department of Industrial and Systems Engineering, The Hong Kong Polytechnic University, Hong Kong SAR

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***Date of birth**: 3 February 2000* | ***Country of citizenship**: People's Republic of China*

Research Profile

I design, build, and empirically evaluate interactive XR systems for multi-user collaboration, multimodal interaction, and remote human-robot interaction. My research combines Unity-based systems prototyping, interaction-technique design, controlled experiments, and mixed-method evaluation to contribute systems and design knowledge for virtual reality, robotics interfaces, multimedia, and visualization.

Research Areas	Human-Computer Interaction; virtual, augmented, and mixed reality; collaborative virtual environments; XR-enabled human-robot interaction; multimodal and haptic interaction; interactive 3D systems; multimedia and visualization; human-centered AI.
Methods	Interactive system design and implementation; Unity/C# prototyping; controlled experiments and dyadic studies; behavioral logging; questionnaire-based evaluation; quantitative and qualitative analysis; systematic reviews.

Education

The Hong Kong Polytechnic University — *Hong Kong SAR*

PhD in Industrial and Systems Engineering, expected May 2027 — *May 2024 - Present*

- Research focus: Human-Computer Interaction, XR collaboration, interactive systems, and XR-enabled human-robot interaction.
- Supervisor: Prof. Lik-Hang Lee & Prof. George Q. Huang

The Hong Kong University of Science and Technology — *Hong Kong SAR*

MPhil in Individualized Interdisciplinary Program - Artificial Intelligence — *Sep. 2021 - Jan. 2024*

- Awarded Postgraduate Studentship Award, 2021-2023.
- Supervisor: Prof. Pan HUI & Prof. Mingming FAN

University of Liverpool — *Liverpool, United Kingdom*

BSc in Electronic Science and Technology, First Class Honours — *Sep. 2017 - Jul. 2021*

Xi'an Jiaotong-Liverpool University — *Suzhou, China*

BSc in Electronic Science and Technology, First Class Honours — *Sep. 2017 - Jul. 2021*

- Supervisor: Prof. Hai-Ning Liang & Prof. Diego Vilela Monteiro

Research and Professional Experience

Hong Kong Productivity Council (HKPC) — Industry Collaboration — *Hong Kong SAR*

Technical Lead — Immersive VR Systems for Industrial Training — *Sep. 2025 - Feb. 2026*

- Led the design and implementation for Unity-based VR training systems for industrial knitting-machine operation, translating industrial procedures into immersive learning scenarios.
- Defined contextual instructions, safety warnings, controller haptics, and machine-state feedback to support guided practice and procedural learning.

The Hong Kong Polytechnic University — *Hong Kong SAR*

Research Assistant – AI-Mediated Immersive Communication — *Jan. 2024 - Apr. 2024*

- Contributed to the development of a research funding proposal on human-centered AI-assisted immersive communication systems for strengthening intimacy, social presence, and shared experiences among geographically separated partners.
- Supported literature synthesis, research framing, and methodological planning, translating ideas from social VR, affective interaction, and AI-mediated communication into a programme of system prototyping and empirical user studies.

Xi'an Jiaotong-Liverpool University — *Suzhou, China*

Undergraduate Researcher – Human-Computer Interaction Lab — *Jun. 2018 - Sep. 2021*

- Built Unity-based VR environments, haptic prototypes, and experimental materials for research on spatial interaction, embodied feedback, and VR training.
- Contributed to system implementation and user-study preparation across multiple collaborative HCI projects.

Suzhou Bio-engin Robot Technology Co. Ltd. — *Suzhou, China*

Research Intern – No. 1 Technical Division — *Jun. 2020 - Sep. 2020*

- Developed data-processing, modeling, and fuzzy-PID-control components for a wire-driven parallel robotic system for gait-rehabilitation training.
- Prepared a technical report on lower-limb control using functional electrical stimulation.

Publications and Patent

First-Author and Equal-Contribution Publications

[1] **Xian Wang**, Xuanru Cheng, Rongkai Shi, Lei Chen, Jingyao Zheng, Hai-Ning Liang, and Lik-Hang Lee. "Conflict Resolution Strategies for Co-manipulation of Virtual Objects under Non-disjoint Conditions." *IEEE Transactions on Visualization and Computer Graphics*, 32(7), 6114-6128, 2026. `https://doi.org/10.1109/TVCG.2026.3677429` **(CCF-A, JCR Q1)**

[2] **Xian Wang**, Luyao Shen, Lei Chen, Mingming Fan, and Lik-Hang Lee. "TeamPortal: Exploring Virtual Reality Collaboration Through Shared and Manipulating Parallel Views." *IEEE Transactions on Visualization and Computer Graphics*, 31(5), 3314-3324, 2025. `https://doi.org/10.1109/TVCG.2025.3549569` **(CCF-A, JCR Q1)**

[3] **Xian Wang**, Luyao Shen, and Lik-Hang Lee. "A Systematic Review of XR-Enabled Remote Human-Robot Interaction Systems." *ACM Computing Surveys*, 57(11), Article 273, 37 pages, 2025. `https://doi.org/10.1145/3730574` **(JCR Q1, IF=30.4)**

[4] **Xian Wang**, Xiaoyu Mo, Lik-Hang Lee, Xiaoying Wei, Xiaofu Jin, Mingming Fan, and Pan Hui. "Designing Loving-Kindness Meditation in Virtual Reality for Long-Distance Romantic Relationships." *Proceedings of the 31st ACM International Conference on Multimedia*, 7608-7617, 2023. `https://doi.org/10.1145/3581783.3612438` **(CCF-A)**

[5] **Xian Wang**, Lik-Hang Lee, Carlos Bermejo Fernandez, and Pan Hui. "The Dark Side of Augmented Reality: Exploring Manipulative Designs in AR." *International Journal of Human-Computer Interaction*, 40(13), 3449-3464, 2024. `https://doi.org/10.1080/10447318.2023.2188799` **(JCR Q1, Citation=76)**

[6] **Xian Wang**, Xiaoyu Mo, Mingming Fan, Lik-Hang Lee, Bertram E. Shi, and Pan Hui. "Reducing Stress and Anxiety in the Metaverse: A Systematic Review of Meditation, Mindfulness and Virtual Reality." *Chinese CHI 2022*, 11 pages, 2022. `https://doi.org/10.1145/3565698.3565781`

[7] **Xian Wang**, Diego Monteiro, Lik-Hang Lee, Pan Hui, and Hai-Ning Liang. "VibroWeight: Simulating Weight and Center of Gravity Changes of Objects in Virtual Reality for Enhanced Realism." *IEEE Haptics Symposium*, 2022. `https://doi.org/10.1109/HAPTICS52432.2022.9765609`

[8] Luyao Shen*, **Xian Wang***, Sijia Li, Lik-Hang Lee, Mingming Fan, and Pan Hui. "EmojiChat: Toward Designing Emoji-Driven Social Interaction in VR Museums." *International Journal of Human-Computer Interaction*, 41(11), 6987-7003, 2025. *Equal contribution. `https://doi.org/10.1080/10447318.2024.2387902` **(JCR Q1)**

Selected Co-Authored Publications

[9] Chi Sun, **Xian Wang**, Abhishek Kumar, Chengbin Cui, and Lik-Hang Lee. "MIRAGE: Multimodal Intention Recognition and Admittance-Guided Enhancement in VR-Based Multi-Object Teleoperation." *IEEE International Symposium on Mixed and Augmented Reality*, 2025. `https://doi.org/10.1109/ISMAR67309.2025.00060`

[10] Jingyao Zheng, **Xian Wang**, Sven Mayer, and Lik-Hang Lee. "Non-urgent Messages Do Not Jump into My Headset Suddenly! Adaptive Notification Design in Mixed Reality." *IEEE Transactions on Visualization and Computer Graphics*, 32(5), 4217-4227, 2026. `https://doi.org/10.1109/TVCG.2026.3679090` **(CCF-A, JCR Q1)**

[11] Jingyao Zheng, Haodi Weng, **Xian Wang**, Chengbin Cui, Sven Mayer, Chi-lok Tai, and Lik-Hang Lee. "PersoNo: Personalised Notification Urgency Classifier in Mixed Reality." *IEEE International Symposium on Mixed and Augmented Reality*, 2025. `https://doi.org/10.48550/arXiv.2508.19622`

[12] Simin Yang, Jiaying Chen, **Xian Wang**, Yang Li, Lik-Hang Lee, and Pan Hui. "SoftHeart: Haptic Heart-Rate Sharing to Enrich Social Cues in Social VR Interaction." *International Journal of Human-Computer Interaction*, published online, 2026. `https://doi.org/10.1080/10447318.2026.2664709` **(JCR Q1)**

[13] Xuanru Cheng, **Xian Wang**, Chi-lok Tai, and Lik-Hang Lee. "MetaRoundWorm: A Virtual Reality Escape Room Game for Learning the Lifecycle and Immune Response to Parasitic Infections." *IEEE Transactions on Visualization and Computer Graphics*, 31(11), 9953-9963, 2025. `https://doi.org/10.1109/TVCG.2025.3616752` **(CCF-A, JCR Q1)**; **2025 Honorable Mention – Best Paper Award**

[14] Xinyu Ning, Yan Zhuo, **Xian Wang**, Chan-In Devin Sio, and Lik-Hang Lee. "When Generative Artificial Intelligence Meets Extended Reality: A Systematic Review." *International Journal of Human-Computer Interaction*, 42(10), 6870-6890, 2026. `https://doi.org/10.1080/10447318.2025.2565392` **(JCR Q1)**

[15] Simin Yang, Yuk Hang Tsui, **Xian Wang**, Ahmad Alhilal, Reza Hadi Mogavi, Xuetong Wang, and Pan Hui. "From Prompt to Metaverse: User Perceptions of Personalized Spaces Crafted by Generative AI." *Companion of the ACM Web Conference*, 2024. `https://doi.org/10.1145/3678884.3681897`

[16] Jingyao Zheng, **Xian Wang**, Simo Hosio, Xiaoxian Xu, and Lik-Hang Lee. "LMLPA: Language Model Linguistic Personality Assessment." *Computational Linguistics*, 51(2), 599-640, 2025. `https://doi.org/10.1162/coli_a_00550` **(JCR Q1)**

[17] Xiaoying Wei, Yizheng Gu, Emily Kuang, **Xian Wang**, Beiyan Cao, Xiaofu Jin, and Mingming Fan. "Bridging the Generational Gap: Exploring How Virtual Reality Supports Remote Communication Between Grandparents and Grandchildren." *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2023. `https://doi.org/10.1145/3544548.3581405` **(CCF-A)**

[18] Xiaofu Jin, Wai Tong, Xiaoying Wei, **Xian Wang**, Emily Kuang, Xiaoyu Mo, Huamin Qu, and Mingming Fan. "Exploring the Opportunity of Augmented Reality in Supporting Older Adults to Explore and Learn Smartphone Applications." *Proceedings of the CHI Conference on Human Factors in Computing Systems*, Article 21, 2024. `https://doi.org/10.1145/3613904.3641901` **(CCF-A)**

[19] Diego Monteiro, Hai-Ning Liang, **Xian Wang**, Wenge Xu, and Huawei Tu. "Design and Development of a Low-cost Device for Weight and Center of Gravity Simulation in Virtual Reality." *ACM International Conference on Multimodal Interaction*, 2021. `https://doi.org/10.1145/3462244.3479907`

[20] Diego Monteiro, **Xian Wang**, Hai-Ning Liang, and Yiyu Cai. "Spatial Knowledge Acquisition in Virtual and Physical Reality: A Comparative Evaluation." *IEEE International Conference on Virtual Reality*, 308-313, 2021. `https://doi.org/10.1109/ICVR51878.2021.9483809`

[21] Diego Monteiro, Hai-Ning Liang, Han Li, Yuchen Fu, and **Xian Wang**. "Evaluating the Need and Effect of an Audience in a Virtual Reality Presentation Training Tool." *Computer Animation and Social Agents*, 62-70, 2020. `https://doi.org/10.1007/978-3-030-63426-1_7`

[22] Diego Monteiro, Hai-Ning Liang, Jinwei Wang, Li Wang, **Xian Wang**, and Yiyu Yue. "Evaluating the Effects of a Cartoon-Like Character with Emotions on Users' Behaviour within Virtual Reality Environments." *IEEE International Conference on Artificial Intelligence and Virtual Reality*, 229-236, 2018. `https://doi.org/10.1109/AIVR.2018.00053`

Patent

[P1] **Patent:** A new device and method for simulating weight and center of gravity in a virtual reality environment. Granted Chinese Patent CN112083803B.

Manuscripts Under Review

[U1] **Xian Wang**, Nieyu Cao, Xuanru Cheng, and Lik-Hang Lee. "SemCollab: Reducing Process Loss in Cross-Scale VR Collaboration through Semantic Zooming" Manuscript under review at *IEEE VR 27*

[U2] **Xian Wang**, Xuanru Cheng, Chi Sun, Yingna Wang, and Lik-Hang Lee. "Supporting Workspace Awareness Across Levels of Scale in Multiscale Collaborative Virtual Environments." Manuscript under review at *IEEE VR 27*

Accepted and In-Press Publications

[A1] Xuanru Cheng, **Xian Wang**, Yue Zhou, Yingna Wang, and Lik-Hang Lee. "Pawspective: Reciprocal Role-Swapping in an Asymmetric VR Game for Cross-Species Empathy." Accepted for publication in the *Proceedings of the 34th ACM International Conference on Multimedia (ACM MM 2026)*, 2026. **(CCF-A)**

[A2] Nieyu Cao, **Xian Wang**, and Lik-Hang Lee. "DP-LENS: A Density-Aware Polyfocal Lens with Topology-Driven Auto-Routing for Occlusion Management in Immersive 3D Analytics." Accepted for publication in *IEEE Transactions on Visualization and Computer Graphics* and for presentation at the *2026 IEEE International Symposium on Mixed and Augmented Reality (IEEE ISMAR 2026)*, 2026. `https://arxiv.org/abs/2607.27697` **(CCF-A, JCR Q1); 2026 Honorable Mention – Best Paper Award**

Research Funding and Grant Development

Hong Kong RGC General Research Fund – Project No. 15201225 *HKD$718,907*

Key Proposal Contributor *2025/26*

- Project: "Mitigate Dark Patterns in Extended Reality E-commerce Configurations: Transitioning from Empirical Research to Computational Evaluation"; Principal Investigator: Prof. Lik-Hang Lee.
- Developed the successful proposal's core research methodology and project plan, encompassing expert-informed speculative design, a 500-participant XR user study, computational analysis of interaction-video data, and an AI-assisted auditing and visualization toolkit.

Entrepreneurship and Industry Leadership

Oryn Tech / Xiangyuan Technology *RMB$2,000,000*

Co-Founder & Chief Operating Officer *2025/26*

- Co-founded an AI hardware venture developing wearable devices and a companion mobile application for personalized, context-aware AI interaction.
- The product concept integrates physiological sensing, user-controlled long-term memory, and haptic feedback to support adaptive interaction in everyday settings.
- Secured seed investment equivalent to RMB 2 million from MiraclePlus.

Teaching and Mentoring Experience

ISE5322: Industrial Metaverse and Mixed Reality *PolyU*

Teaching Assistant *2024-2025*

- Developed approximately 10 hours of instructional videos on virtual reality and Unity-based immersive-system development.
- Designed and delivered a two-hour class on participatory design, including a hands-on workshop exercise.
- Guided students in a course project examining generative AI and extended reality, and subsequently contributed to its development into a co-authored systematic review published in the *International Journal of Human-Computer Interaction* (JCR Q1).

CMAA 5017: AR/VR/MR/XR: Concepts, Theory and Techniques *HKUST*

Teaching Assistant *2022-2023*

- Prepared and delivered a two-hour lecture on virtual-reality concepts, interaction techniques, and immersive-system applications.
- Supported course assessment, grading, and examination administration.

ISE4025: Enterprise Analytics; ISE526: Enterprise Resources Planning *PolyU*

Teaching Assistant *2025-2026*

- Supported grading, examinations, and other assessment activities across both courses.
- Coordinated guest-speaker arrangements and related course activities.

Ready to Teach	Human–Computer Interaction; Virtual and Augmented Reality; Interactive 3D Systems; Multimedia Computing; Human–Robot Interaction; Research Methods for Interactive Systems.
Teaching Strengths	Technical content preparation; VR/Unity instruction; participatory design workshop facilitation; assessment and invigilation; guest speaker coordination.

Selected Interactive Systems and Research Projects

Conflict Resolution for VR Co-Manipulation *PolyU*

Project Lead and First Author *2025-2026*

- Designed and evaluated preventive and reactive strategies for multi-user manipulation of overlapping virtual object sub-components.
- Conducted two dyadic user studies with 76 participants, producing design implications for flexible VR collaboration systems.

TeamPortal: Shared and Manipulable Perspectives for VR Collaboration *PolyU*

Project Lead and First Author *2024-2025*

- Built a VR collaboration system that lets collaborators share and manipulate each other's field of view through interactive view windows.
- Ran two studies with 72 participants, showing improvements in efficiency, movement reduction, social presence, and adoption willingness in complex tasks.

XR-Enabled Remote Human-Robot Interaction *PolyU*

Project Lead and First Author *2024-2025*

- Led a systematic review of 100 XR-enabled remote HRI systems selected from an initial corpus of 2,561 articles using a PRISMA-informed process.
- Identified interaction design gaps in multimodal feedback, user-centered teleoperation, and practical deployment of XR interfaces for robots.

Professional Service

2025	**Secretary**, ISE Research Students' Conference 2025	*PolyU*
2022	**Program Committee Member**, IEEE International Symposium on Mixed and Augmented Reality	*Singapore*
2023	**Web Chair**, ACM MobiSys	
2022-2026	**Reviewer**, CHI, CSCW, IEEE VR, ISMAR, and ACM Multimedia	
2023	**Reviewer**, Cyberpsychology, Behavior, and Social Networking	
2024	**Reviewer**, IEEE Transactions on Affective Computing	

Honors, Awards, and Leadership

2026	**Co-recipient**, 2026 Honorable Mention – Best Paper Award, IEEE International Symposium on Mixed and Augmented Reality (ISMAR)	*Honorable Mention – Best Paper Award*
2026	**Co-founder & COO**, Oryn Tech / Xiangyuan Technology; secured RMB 2M-equivalent seed investment from MiraclePlus	*AI hardware startup*
2025	**Co-recipient**, 2025 Honorable Mention – Best Paper Award, IEEE International Symposium on Mixed and Augmented Reality (ISMAR)	*Honorable Mention – Best Paper Award*
2025	**Top Performer, Micro Fund**, HKD 20,000	*PolyU*
2023	**Research Travel Grant**, HKD 13,500	*HKUST*
2018-2021	**President**, Xi'an Jiaotong-Liverpool University Student-Staff Liaison Committee	*XJTLU*
2019	**Third Prize**, RoboMaster 2019 International Regional Competition	

Technical Skills

XR Development	Unity, C#, VR/AR/MR prototyping, interactive 3D environments, embodied interaction systems, VR haptic prototypes, game-engine experimental platforms.
Programming	Python, C/C++, C#, MATLAB, JavaScript, Verilog, LaTeX, Arduino-based prototyping, data analysis scripts.
Design Tools	3ds Max, Photoshop, prototype visualization, storyboard-style project documentation, 3D scanning workflows.
Hardware	VR head-mounted displays and controllers, haptic devices, liquid-metal/fluidic prototypes, OpenBCI, Arduino, sensors, TwinCAT.
Languages	Mandarin Chinese, English.